

Windows 7

Windows 7, the successor to Windows Vista, will incorporate virtualisation technology called XP mode for compatibility with older programs

Vendors bet big

Mobile Virtualisation firm VirtualLogix is funded by Cisco, Intel Capital, Motorola and Texas Instruments, all big names in the technology business

How stuff works

system. The same argument holds true for users as well since they can use an application developed for a different platform on their smartphones using mobile virtualisation and a corresponding virtual machine.

While we have just seen many advantages of mobile virtualisation above, some challenges lie ahead for this technology as well.

Under mobile virtualisation, each virtual machine has its own scheduling processes and tasks running in the background. Unlike a single operating system, multiple virtual machines will not share data regarding scheduling requirements for different tasks and processes, etc. This can result in degradation of performance across all virtual machines running on that device.

Mobile virtualisation suffers from some performance and battery life issues. There is a performance penalty since all commands to the mobile hardware must pass through the hypervisor which adds an extra layer of waiting time since multiple virtual machines are attempting to use common hardware. So the designers have to ensure that performance does not drop noticeably when multiple virtual machines are started on the phone hardware. Most smartphones are tweaked for battery life as part of a process that involves significant interaction between the operating system, device drivers and running applications. However, since each virtual machine functions in its own silo, none of them may be able to make decisions regarding energy management.

While time to market for the manufacturer decreases since they do not need to tweak the operating system each time, the hypervisor needs to be able to support the many different hardware combinations which may be used in the smartphone market. Companies like ARM license their chip designs to mobile companies who may make their own design modifications to the processors. In order to support that particular hardware, the chip design and modifications must all be verified

by the company supporting the virtualisation hypervisor.

The mobile virtualisation hypervisor must be provided by the handset manufacturer as a built in feature when shipping the product. As a result, it becomes critical for the virtualisation companies to partner with handset manufacturers and ensure that enough partners are on board to support and use their technology.

Now that we understand the technology, we can take a look at the major companies which are working on mobile virtualisation.

The largest firm in this group is VMware Inc., owned by storage major EMC. VMware was one of the early pioneers in virtualisation and is a global leader in the virtualisation space. Though it is not available to manufacturers yet, VMware entered the mobile virtualisation market with the VMware Mobile Virtualisation Platform (MVP) which they acquired from Trango Virtual Processors in November 2008. The MVP software contains a hypervisor and supports mobile virtualisation on ARM processors and multiple smartphone operating systems. VMware demoed a Nokia N800 running Android along with Windows CE at VMworld Europe in February 2009.

Open Kernel Labs is an existing provider of mobile virtualisation solutions for multiple smartphone operating systems. It has partnered with heavyweights like Motorola, Citrix, ARM and a number of handset manufacturers for incorporating mobile virtualisation into their products. It has already launched a phone based on mobile virtualisation called the 'Motorola Evoke QA4'.

VirtualLogix is another company in the mobile virtualisation field. Cisco, Intel, Motorola and Texas Instruments (TI) hold a stake in this company for development of products based on mobile virtualisation.

Xen is an open source project in the virtualisation field which has also released support for mobile virtualisation using

ARM's ARM 9 processor family. Samsung is developing cell phones based on mobile virtualisation and the Xen project.

Research In Motion (RIM), the maker of the popular Blackberry software, have also announced that they will release a version of their blackberry operating system which will run on Windows Mobile phones. This is also based on Multi-OS mobile virtualisation.

While VMware is a major player in the virtualisation space, it does not have a smartphone based on the Mobile Virtualisation Platform in the market yet. Both of its major competitors, Open Kernel Labs and VirtualLogix have already released products for mobile virtualisation and are backed by large hardware companies who will benefit when mobile virtualisation takes off. So these two firms may be targets for an acquisition once the mobile virtualisation market develops further. The Xen open source project also has substantial technical expertise behind it. It was responsible for the drop in prices of server hypervisor software since open source software is free of cost. Hence, it too cannot be underestimated.

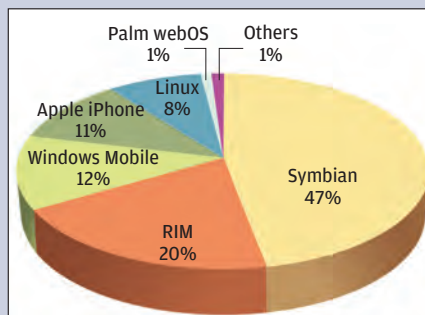
According to Gartner research, the mobile virtualisation segment is expected to grow at a fast pace by the year 2012. However, the current market is growing quite slowly. While some of this can be attributed to the effects of the worldwide economic recession and reduced spending on IT, the fact remains that server virtualisation also took some time to become prevalent in datacenters. The positive sign is the entry of established virtualisation companies like VMware into this field because this shows that they are also expecting this market to grow in the future. Considering these facts, mobile virtualisation companies have sufficient time to work on its negatives. On the other hand, looking at the benefits that users get from mobile virtualisation, we can safely say that mobile virtualisation is going to be mainstream one day. **d**

SMART OS

There are seven smartphone operating systems that are widely used in the market today:

- **Symbian OS:** Symbian, owned by Nokia, has the largest share in the worldwide smartphone market although its market share is very small in North America. Although Symbian is used by other handset manufacturers as well, each of them uses its own implementation of Symbian due to which all applications are incompatible across the different versions.
- **RIM BlackBerry:** Research In Motion's Blackberry is a well-known business phone known for its strong email and productivity applications. RIM is soon going to release a version of Blackberry that will run on Windows Mobile as a virtual machine.

- **Windows Mobile:** Windows Mobile is Microsoft's smartphone operating system, currently at version 6.5. Substantial improvements and better touch screen



The pie chart shows sales of smartphones as of Q4 2008. Source: Gartner Research

support are expected in the upcoming Version 7.

- **iPhone OS:** This is a lighter version of Apple's Mac OS X. The latest version is 3.0.
- **Linux:** Linux is an open source operating system. Many companies, including Motorola, use it in their smartphones after changing the source code as per their requirements. Linux is available free of cost.
- **Palm webOS:** Palm has been losing market share over the last few years. The new Palm device, the Palm Pre has gotten good reviews as an iPhone challenger.
- **Android:** As of Q2 2009, Google's Android had a 2.2% share of the worldwide smartphone market (Source: Gartner). It is based on Linux and is an open source operating system.